

YIHENG SU

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EDUCATION

University of Wisconsin-Madison

PhD in Computer Science; **GPA: 3.875/4.00**

Madison, WI

Aug. 2024 – Present

Colby College

Bachelor of Arts in Mathematics and Computer Science; **GPA: 4.00/4.00**

Waterville, ME

Sep. 2020 – May 2024

Honors: Marston Morse Prize in Mathematics, Dean's List (SP22, FA22, FA23, SP24)

PUBLICATIONS

- Yiheng Su, Emmanouil-Vasileios Vlatakis-Gkaragkounis. *On the Exploitability of FTRL Dynamics*. Under review.
- Yiheng Su, Emmanouil-Vasileios Vlatakis-Gkaragkounis, Pucheng Xiong. *Breaking $1/\epsilon$ Barrier in Quantum Zero-Sum Games: Generalizing Metric Subregularity for Spectraplexes*. Under review.
- Peter Engel, Owen Hammond-Lee, Yiheng Su, Dániel Varga, and Pál Zsámboki. *Diverse Beam Search to Find Densest-Known Planar Unit Distance Graphs*. Experimental Mathematics, 2025.
- Bernard Akwei, Bobita Atkins, Rachel Bailey, Ashka Dalal, Natalie Dinin, Jonathan Kerby-White, Tess McGuinness, Tonya Patricks, Luke Rogers, Genevieve Romanelli, Yiheng Su, and Alexander Teplyaev. *Convergence, Optimization, and Stability of Singular Eigenmaps*. Preprint, 2024.
- Hannen Wolfe, Yiheng Su, and Jue Wang. *Dimensional Design of Emotive Sounds for Robots*. HRI 2024.

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Wisconsin, Madison

Madison, WI

Advisor: Emmanouil-Vasileios Vlatakis-Gkaragkounis

Aug. 2024 – Present

- Studied the exploitability of Follow-the-Regularized-Leader in two-player zero-sum matrix games, showing that adversarial surplus is inherent rather than implementation-specific by deriving tight lower bounds against clairvoyant and alternating optimizers.
- Designed quantum min-max algorithms for quantum zero-sum games achieving linear last-iterate convergence ($O(\log(1/\epsilon))$ iterations), improving over the $O(1/\epsilon)$ average-iterate rate of Vasconcelos et al. and yielding an exponential speedup over the Jain-Watrous method for parallel approximation of strictly positive SDPs.

Research Assistant, Alfréd Rényi Institute of Mathematics

Budapest, Hungary

Supervisor: Pál Zsámboki

Feb. 2023 – Aug. 2024

- Developed a beam-search algorithm with a visitation metric to construct dense unit distance graphs, recovering all known maximally dense configurations up to isomorphism and extending computational bounds to $n \leq 100$.
- Generated a database of over 60 million graphs and provided new evidence on how the maximum number of edges scales with the number of vertices.

National Science Foundation REU Researcher, University of Connecticut

Storrs, CT

Supervisors: Luke Rogers and Alexander Teplyaev

May 2023 – Aug. 2023

- Studied the approximation of Laplacian eigenmaps in nonlinear dimensionality reduction, identifying empirically optimal scaling regimes for the kernel bandwidth ϵ through solvable models and Monte Carlo simulations.
- Validated accuracy across domains including intervals, manifolds (torus, sphere), and fractal spaces, providing practical guidance for parameter selection in spectral methods.

Research Assistant, Colby College

Waterville, ME

Supervisor: Hannen Wolfe

May 2022 – Apr. 2023

- Investigated non-linguistic utterances for emotional expression in human-robot interaction, developing a mapping from musical and prosodic features to a pleasure-arousal emotion model.
- Validated the design through user studies, showing reliable differentiation of excitement, sadness, and anger, while highlighting limitations in conveying contentment.

TEACHING EXPERIENCE

Teaching Assistant, University of Wisconsin–Madison <i>CS300 Programming II (FA24, SP25, SU25, FA25, SP26)</i>	Madison, WI <i>Aug. 2024 – Present</i>
Teaching Assistant, Colby College <i>Abstract Algebra (FA23), ODE (SP22), Algorithms (FA22, SP22), Vector Calculus (FA22)</i>	Waterville, ME <i>Sept. 2021 – May 2024</i>
Volunteer Math Teacher, Albert S. Hall School <i>5-th grade Mathematics: Number and Operations</i>	Waterville, ME <i>Feb. 2022 – May 2023</i>

SKILLS & INTERESTS

- Programming:** Python, Java, R, MATLAB, Mathematica
- Tool & Frameworks:** NumPy, pandas, scikit-learn, TensorFlow, SQL, GitHub, LaTeX
- Languages:** English, Native Chinese (Mandarin)
- Interests:** Badminton, Ukulele